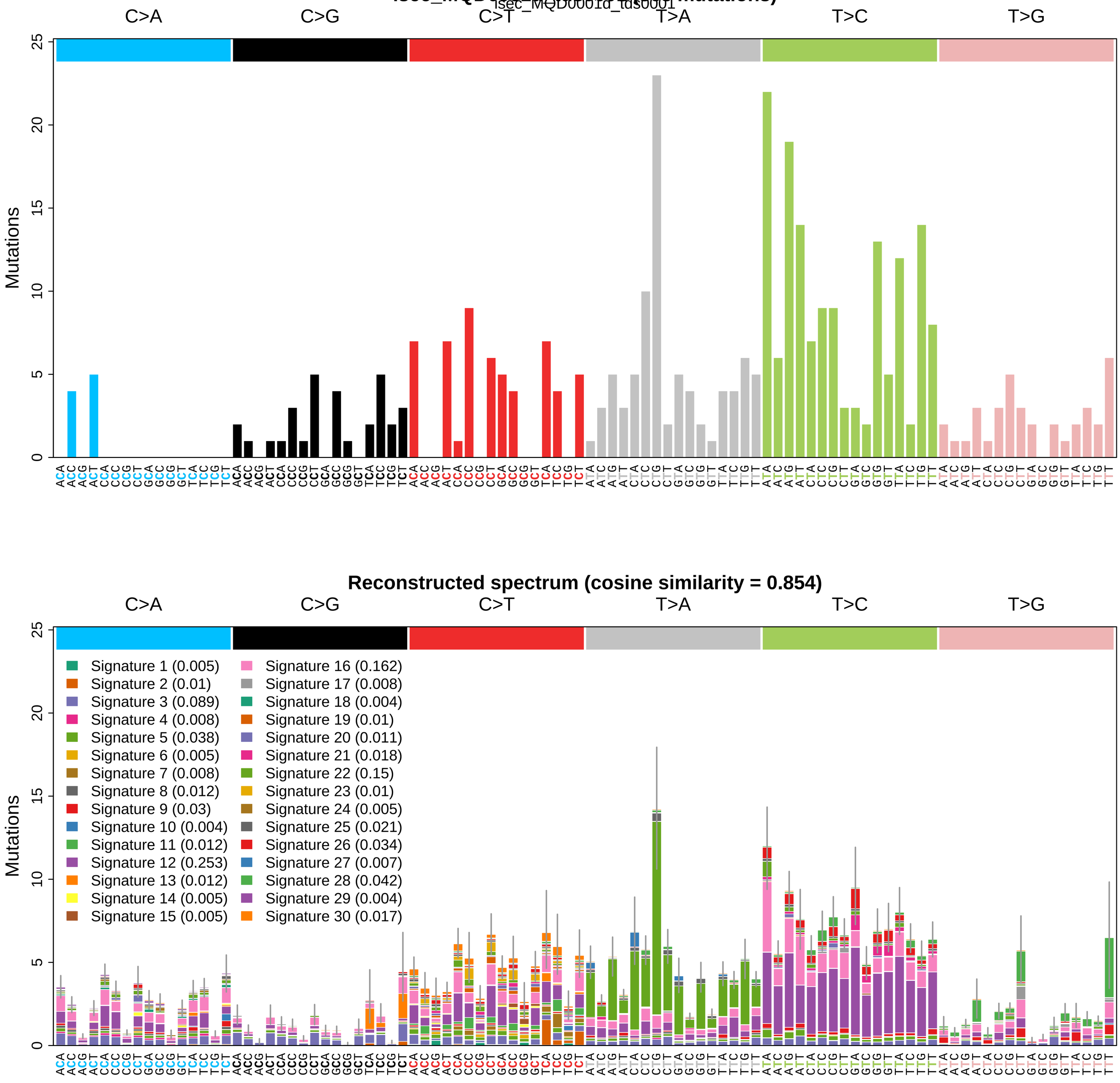
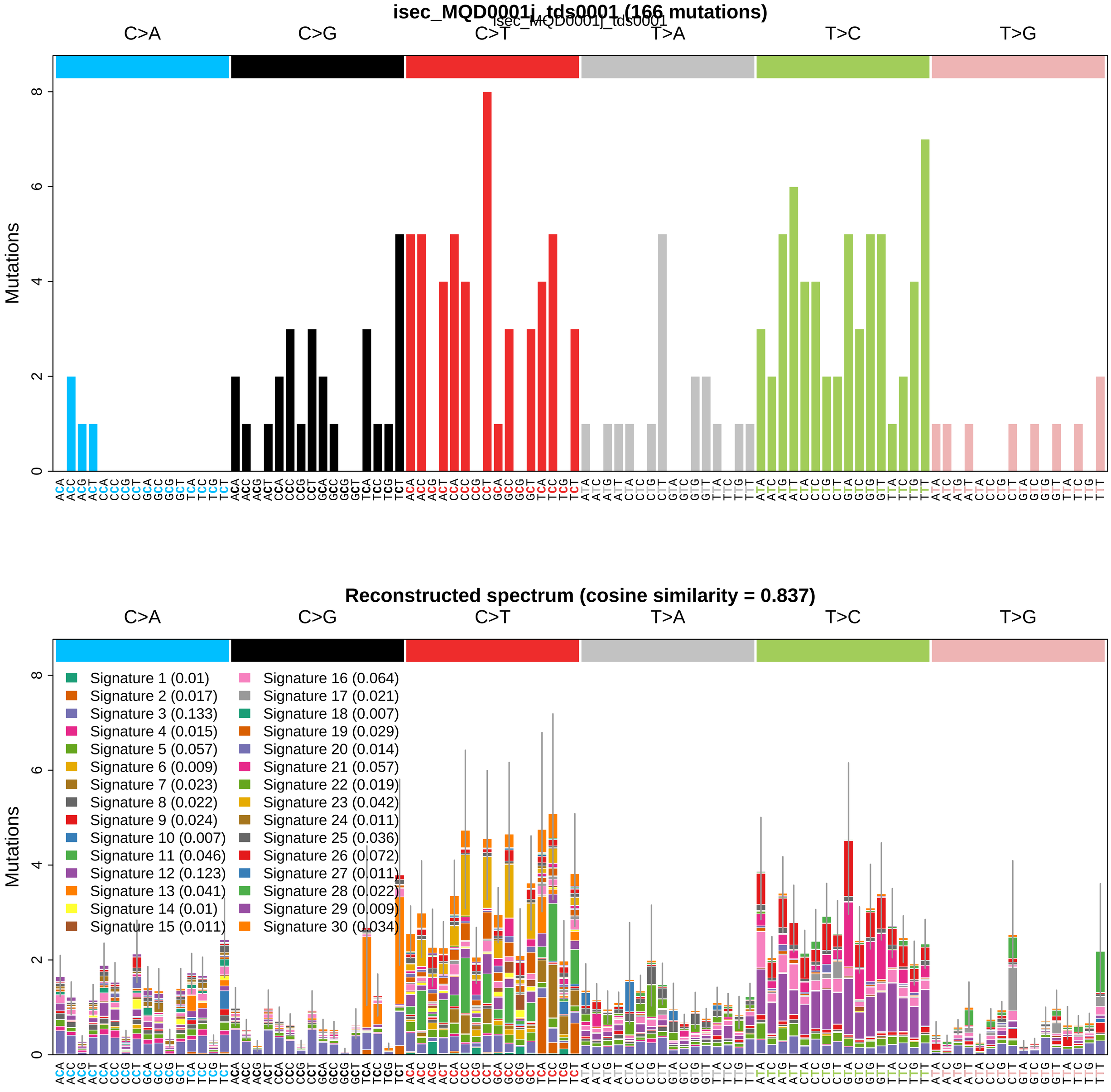
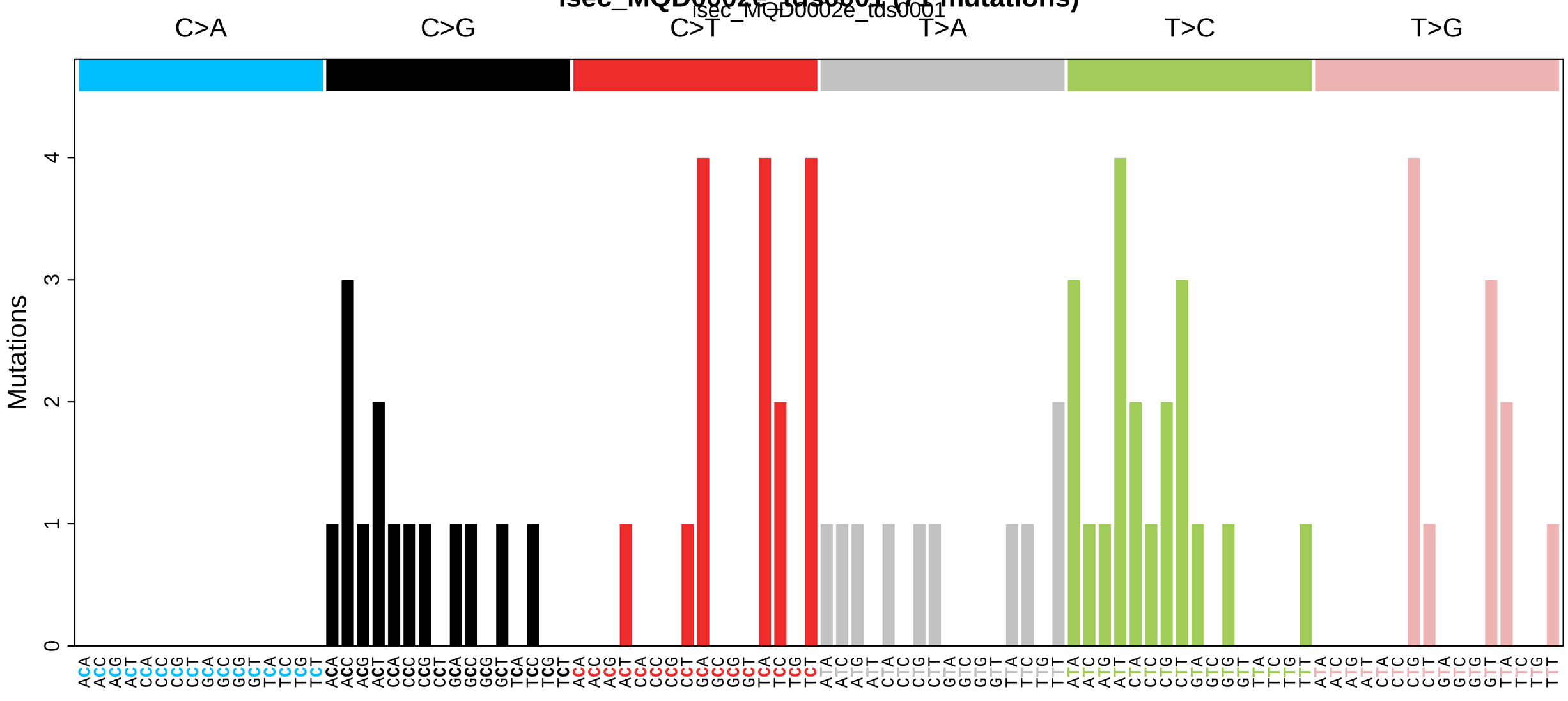
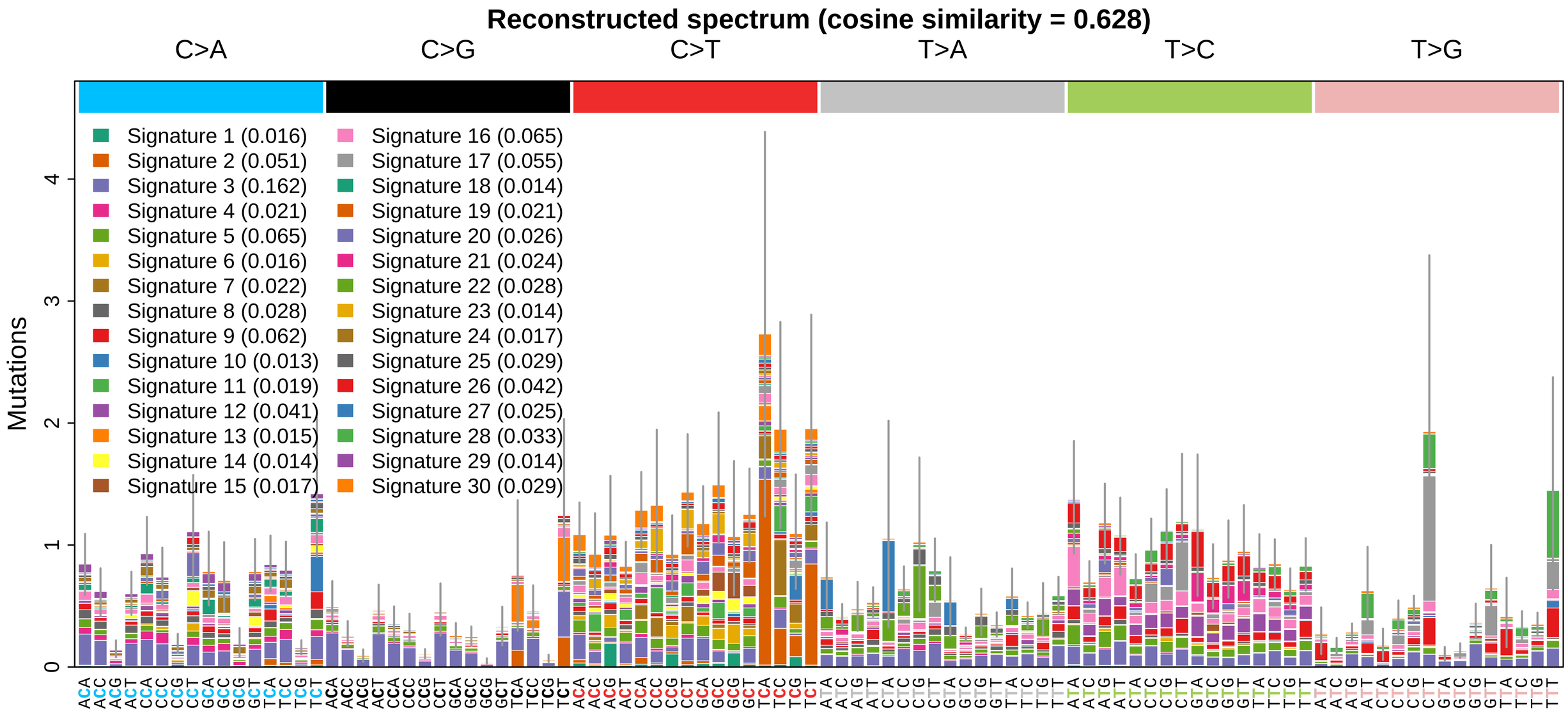
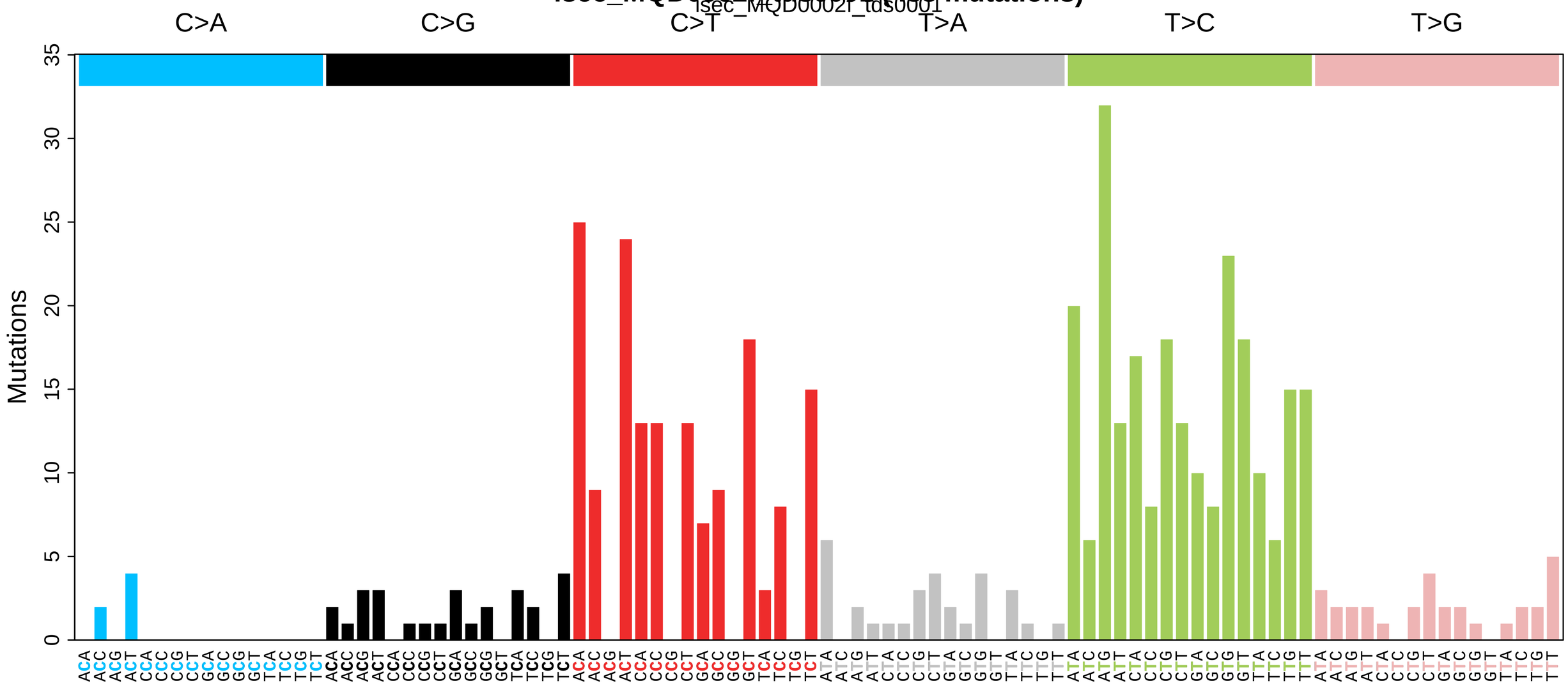
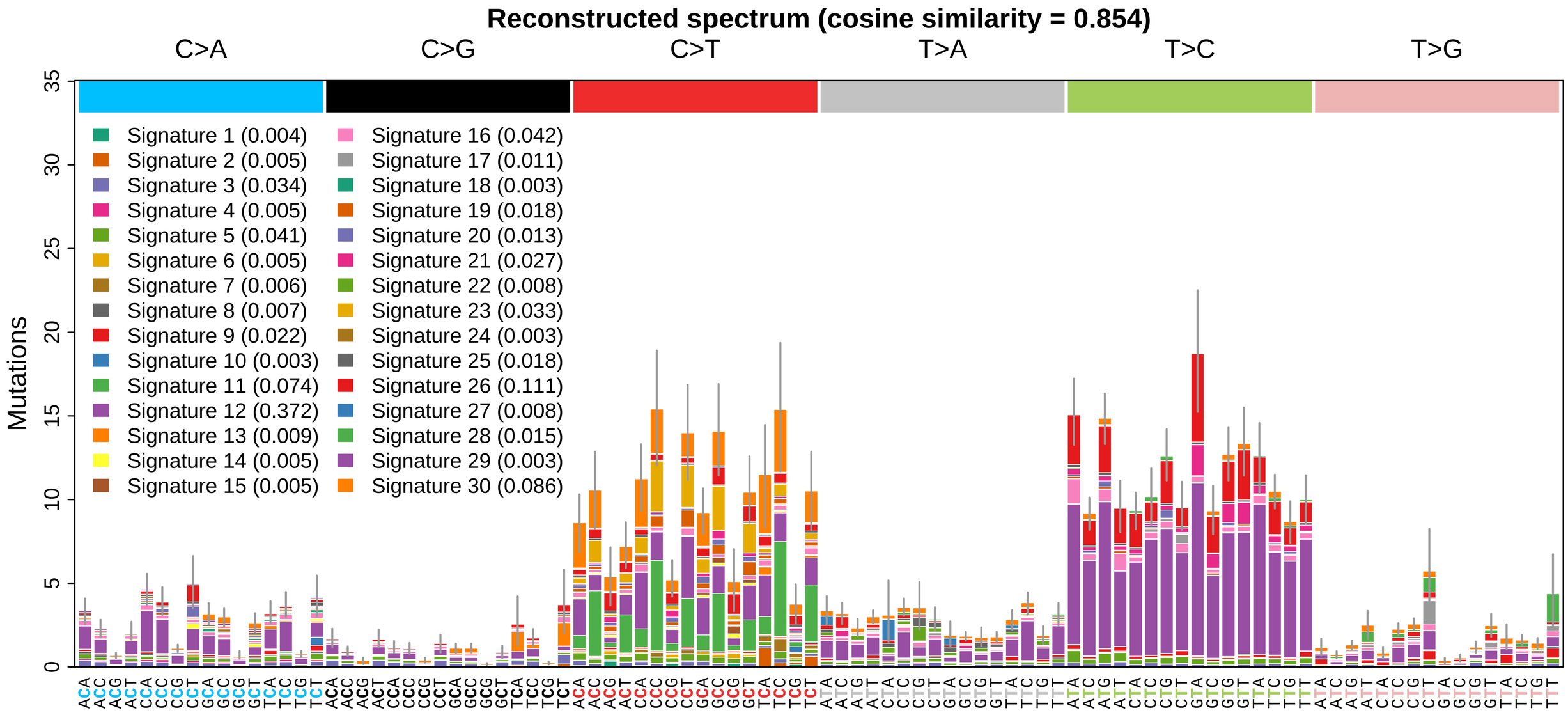

isec_MQD0001d_tds0001 (363 mutations)
isec_MQD0001d_tds0001
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
5
10
15
20
25
Reconstructed spectrum (cosine similarity = 0.854)
Signature 1 (0.005)
Signature 2 (0.01)
Signature 3 (0.089)
Signature 4 (0.008)
Signature 5 (0.038)
Signature 6 (0.005)
Signature 7 (0.008)
Signature 8 (0.012)
Signature 9 (0.03)
Signature 10 (0.004)
Signature 11 (0.012)
Signature 12 (0.253)
Signature 13 (0.012)
Signature 14 (0.005)
Signature 15 (0.005)
Signature 16 (0.162)
Signature 17 (0.008)
Signature 18 (0.004)
Signature 19 (0.01)
Signature 20 (0.011)
Signature 21 (0.018)
Signature 22 (0.15)
Signature 23 (0.01)
Signature 24 (0.005)
Signature 25 (0.021)
Signature 26 (0.034)
Signature 27 (0.007)
Signature 28 (0.042)
Signature 29 (0.004)
Signature 30 (0.017)

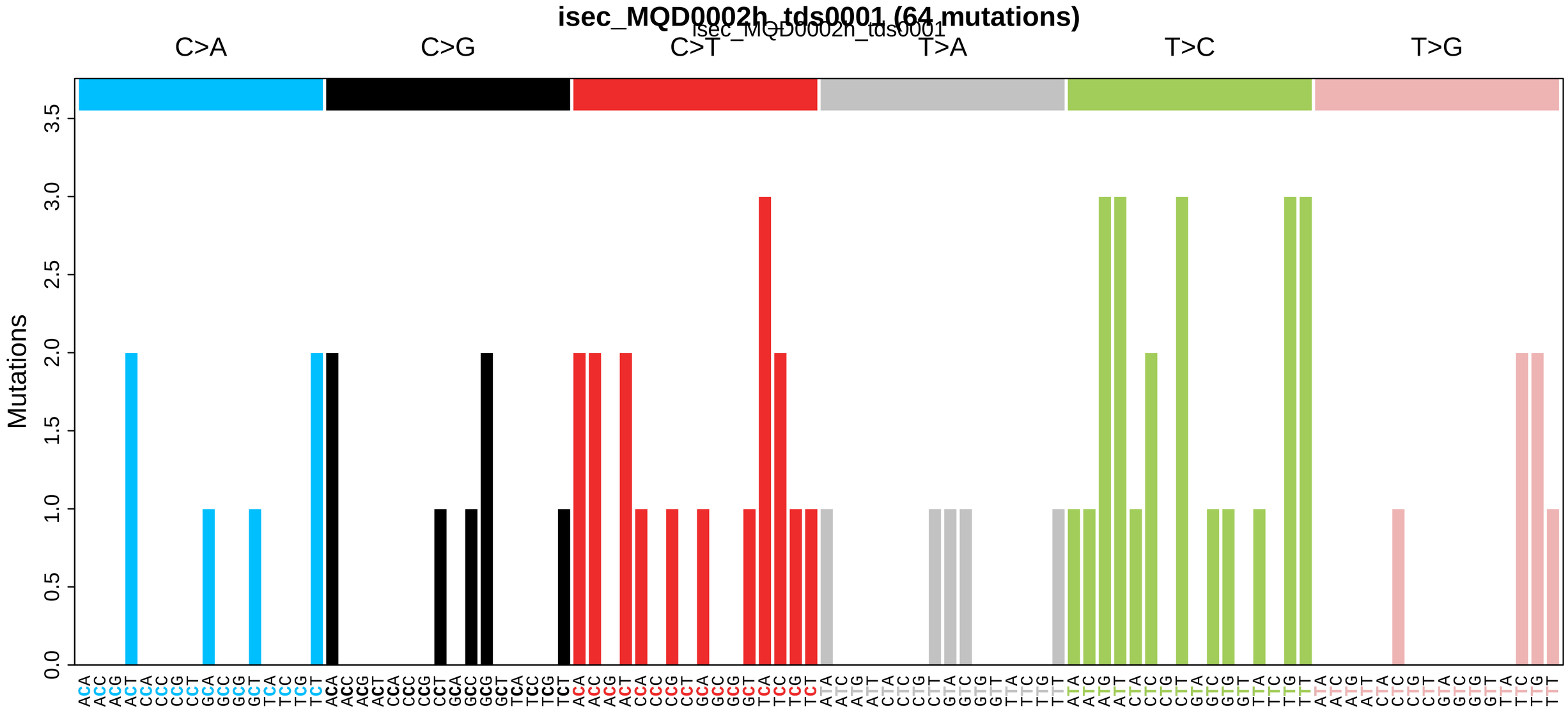

isec_MQD0001j_tds0001 (166 mutations)
isec_MQD0001j_tds0001
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
2
4
6
8
Reconstructed spectrum (cosine similarity = 0.837)
Signature 1 (0.01)
Signature 2 (0.017)
Signature 3 (0.133)
Signature 4 (0.015)
Signature 5 (0.057)
Signature 6 (0.009)
Signature 7 (0.023)
Signature 8 (0.022)
Signature 9 (0.024)
Signature 10 (0.007)
Signature 11 (0.046)
Signature 12 (0.123)
Signature 13 (0.041)
Signature 14 (0.01)
Signature 15 (0.011)
Signature 16 (0.064)
Signature 17 (0.021)
Signature 18 (0.007)
Signature 19 (0.029)
Signature 20 (0.014)
Signature 21 (0.057)
Signature 22 (0.019)
Signature 23 (0.042)
Signature 24 (0.011)
Signature 25 (0.036)
Signature 26 (0.072)
Signature 27 (0.011)
Signature 28 (0.022)
Signature 29 (0.009)
Signature 30 (0.034)

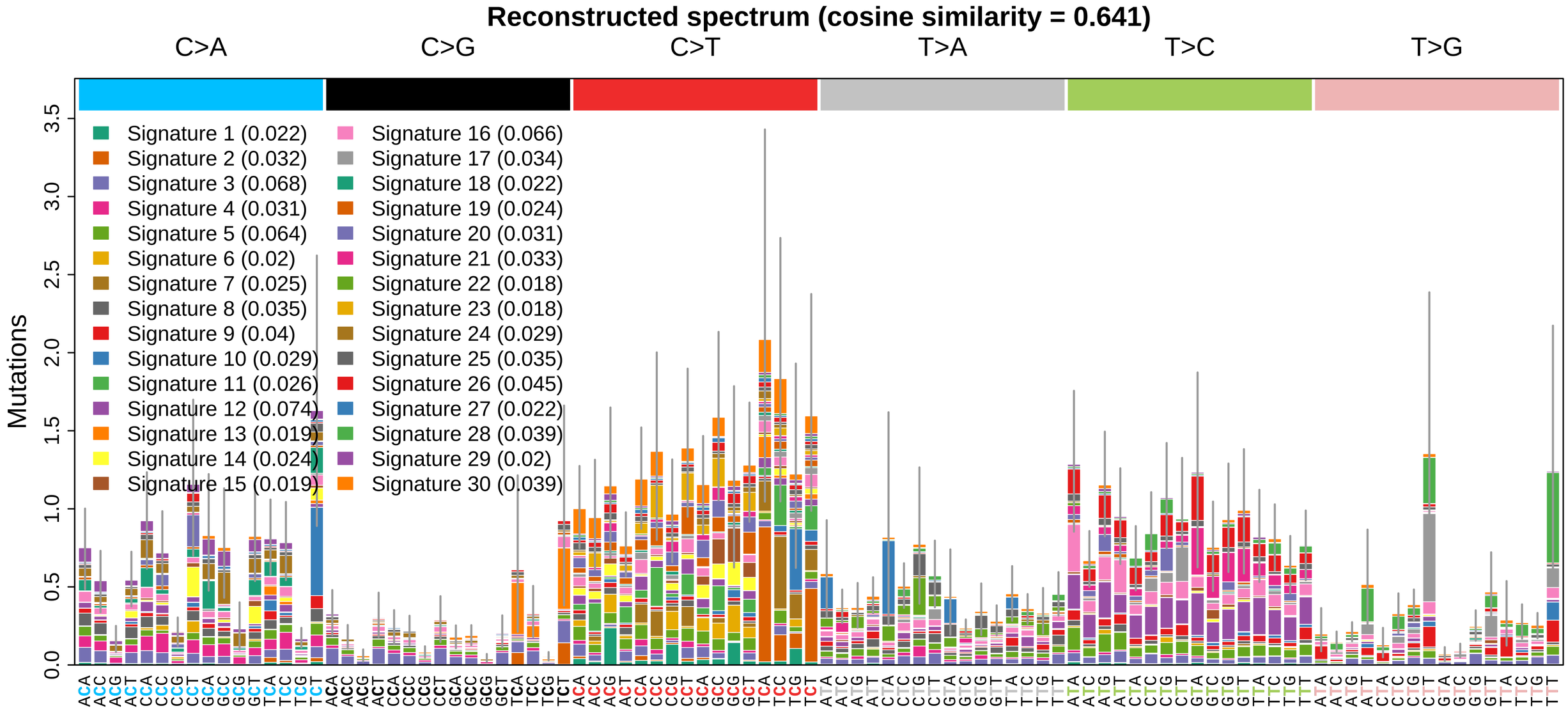

isec_MQD0002e_tds0001 (71 mutations)
isec_MQD0002e_tds0001
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
1
2
3
4

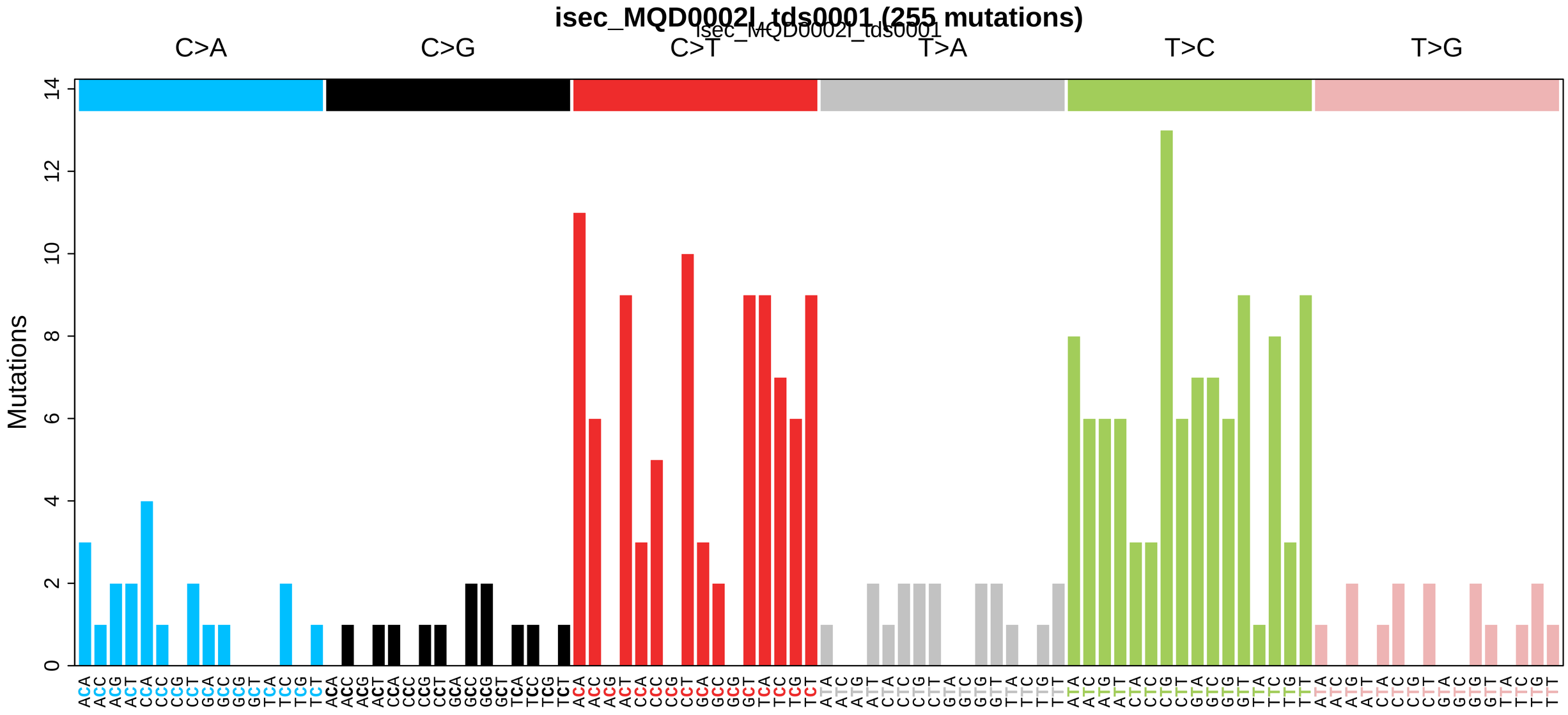

Reconstructed spectrum (cosine similarity = 0.628)
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
1
2
3
4
Signature 1 (0.016)
Signature 2 (0.051)
Signature 3 (0.162)
Signature 4 (0.021)
Signature 5 (0.065)
Signature 6 (0.016)
Signature 7 (0.022)
Signature 8 (0.028)
Signature 9 (0.062)
Signature 10 (0.013)
Signature 11 (0.019)
Signature 12 (0.041)
Signature 13 (0.015)
Signature 14 (0.014)
Signature 15 (0.017)
Signature 16 (0.065)
Signature 17 (0.055)
Signature 18 (0.014)
Signature 19 (0.021)
Signature 20 (0.026)
Signature 21 (0.024)
Signature 22 (0.028)
Signature 23 (0.014)
Signature 24 (0.017)
Signature 25 (0.029)
Signature 26 (0.042)
Signature 27 (0.025)
Signature 28 (0.033)
Signature 29 (0.014)
Signature 30 (0.029)

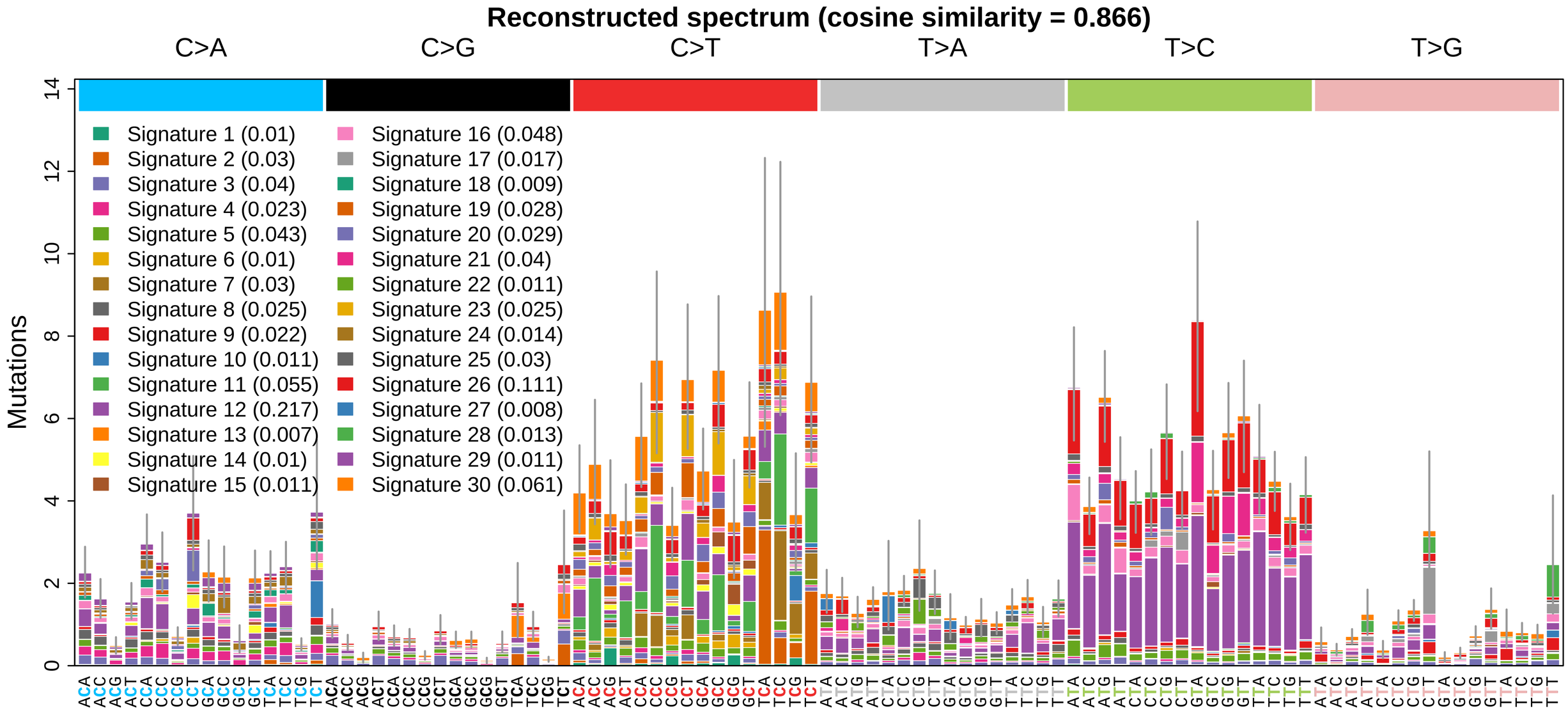

isec_MQD0002f_tds0001 (483 mutations)
isec_MQD0002f_tds0001
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
5
10
15
20
25
30
35

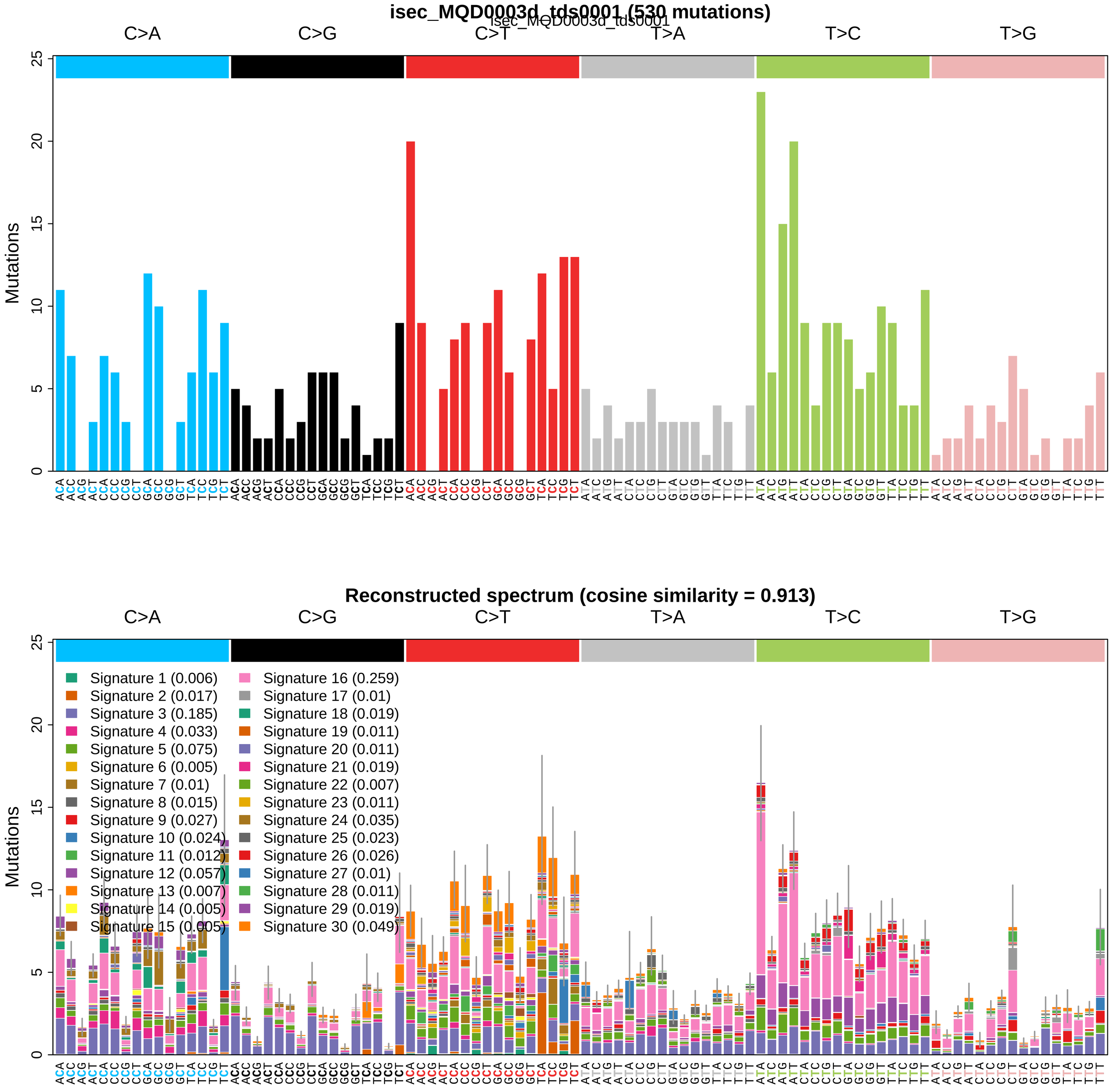

Reconstructed spectrum (cosine similarity = 0.854)
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
5
10
15
20
25
30
35
Signature 1 (0.004)
Signature 2 (0.005)
Signature 3 (0.034)
Signature 4 (0.005)
Signature 5 (0.041)
Signature 6 (0.005)
Signature 7 (0.006)
Signature 8 (0.007)
Signature 9 (0.022)
Signature 10 (0.003)
Signature 11 (0.074)
Signature 12 (0.372)
Signature 13 (0.009)
Signature 14 (0.005)
Signature 15 (0.005)
Signature 16 (0.042)
Signature 17 (0.011)
Signature 18 (0.003)
Signature 19 (0.018)
Signature 20 (0.013)
Signature 21 (0.027)
Signature 22 (0.008)
Signature 23 (0.033)
Signature 24 (0.003)
Signature 25 (0.018)
Signature 26 (0.111)
Signature 27 (0.008)
Signature 28 (0.015)
Signature 29 (0.003)
Signature 30 (0.086)

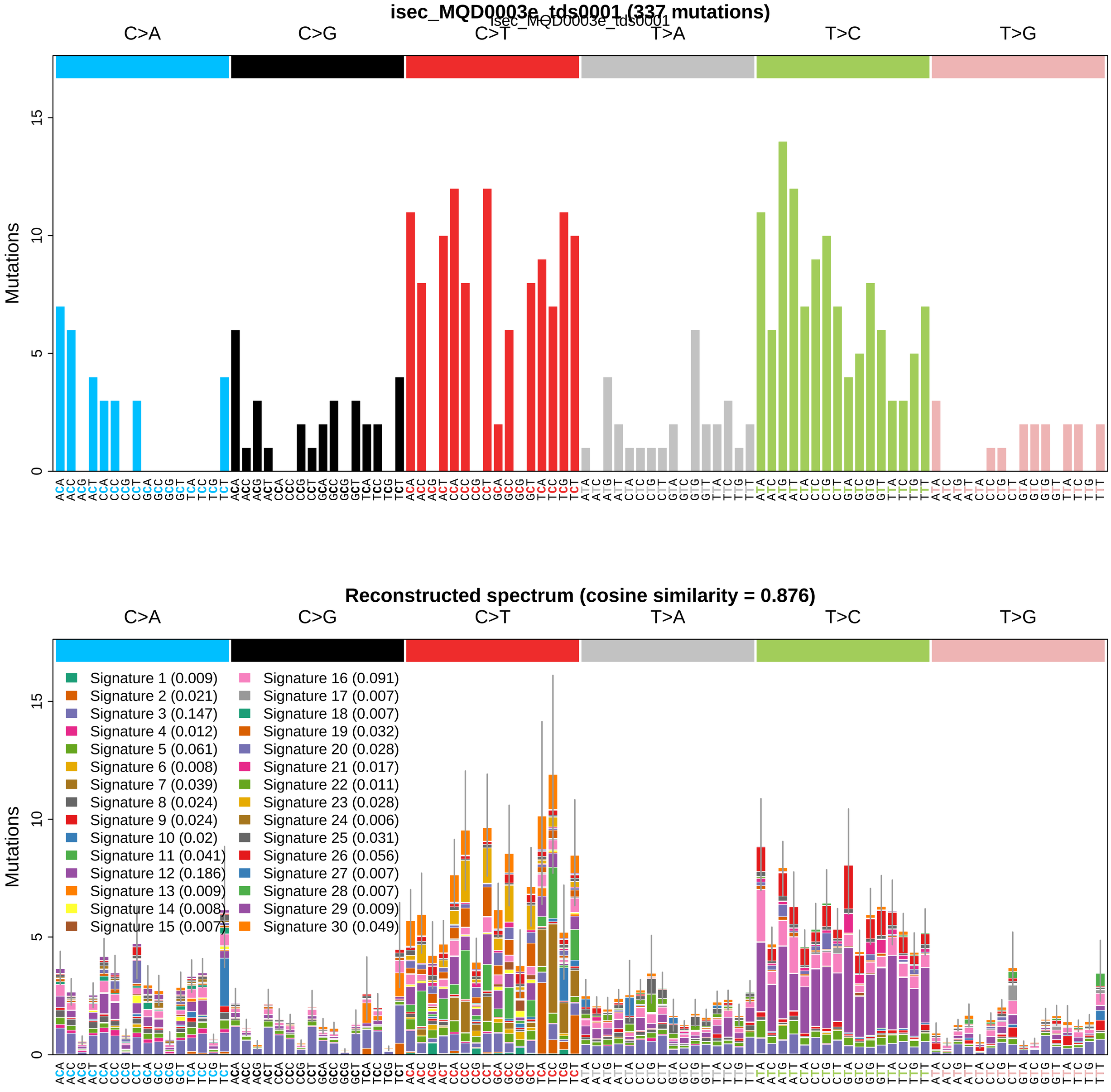

isec_MQD0002h_tds0001 (64 mutations)
isec_MQD0002h_tds0001
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5

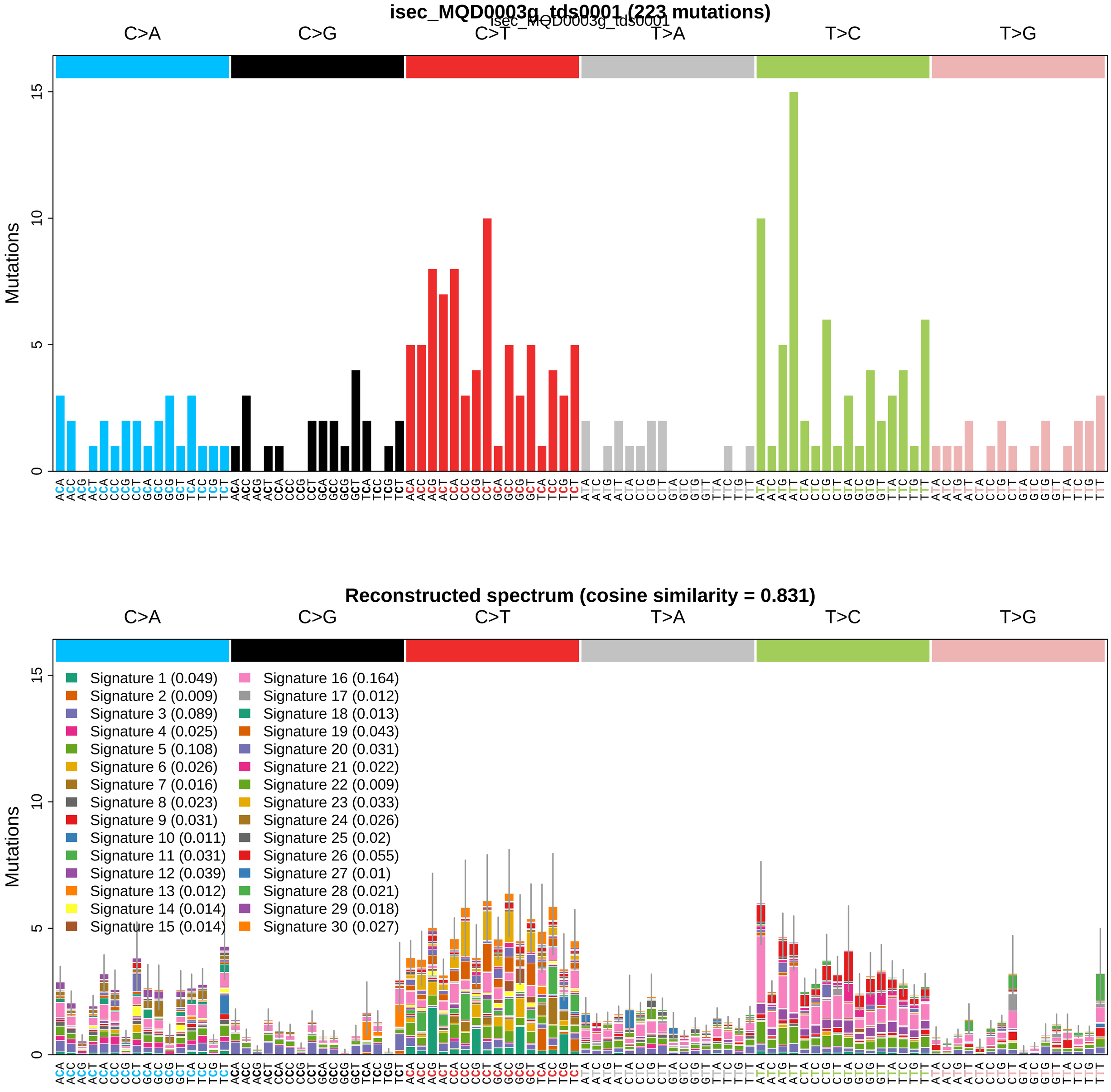

Reconstructed spectrum (cosine similarity = 0.641)
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Signature 1 (0.022)
Signature 2 (0.032)
Signature 3 (0.068)
Signature 4 (0.031)
Signature 5 (0.064)
Signature 6 (0.02)
Signature 7 (0.025)
Signature 8 (0.035)
Signature 9 (0.04)
Signature 10 (0.029)
Signature 11 (0.026)
Signature 12 (0.074)
Signature 13 (0.019)
Signature 14 (0.024)
Signature 15 (0.019)
Signature 16 (0.066)
Signature 17 (0.034)
Signature 18 (0.022)
Signature 19 (0.024)
Signature 20 (0.031)
Signature 21 (0.033)
Signature 22 (0.018)
Signature 23 (0.018)
Signature 24 (0.029)
Signature 25 (0.035)
Signature 26 (0.045)
Signature 27 (0.022)
Signature 28 (0.039)
Signature 29 (0.02)
Signature 30 (0.039)

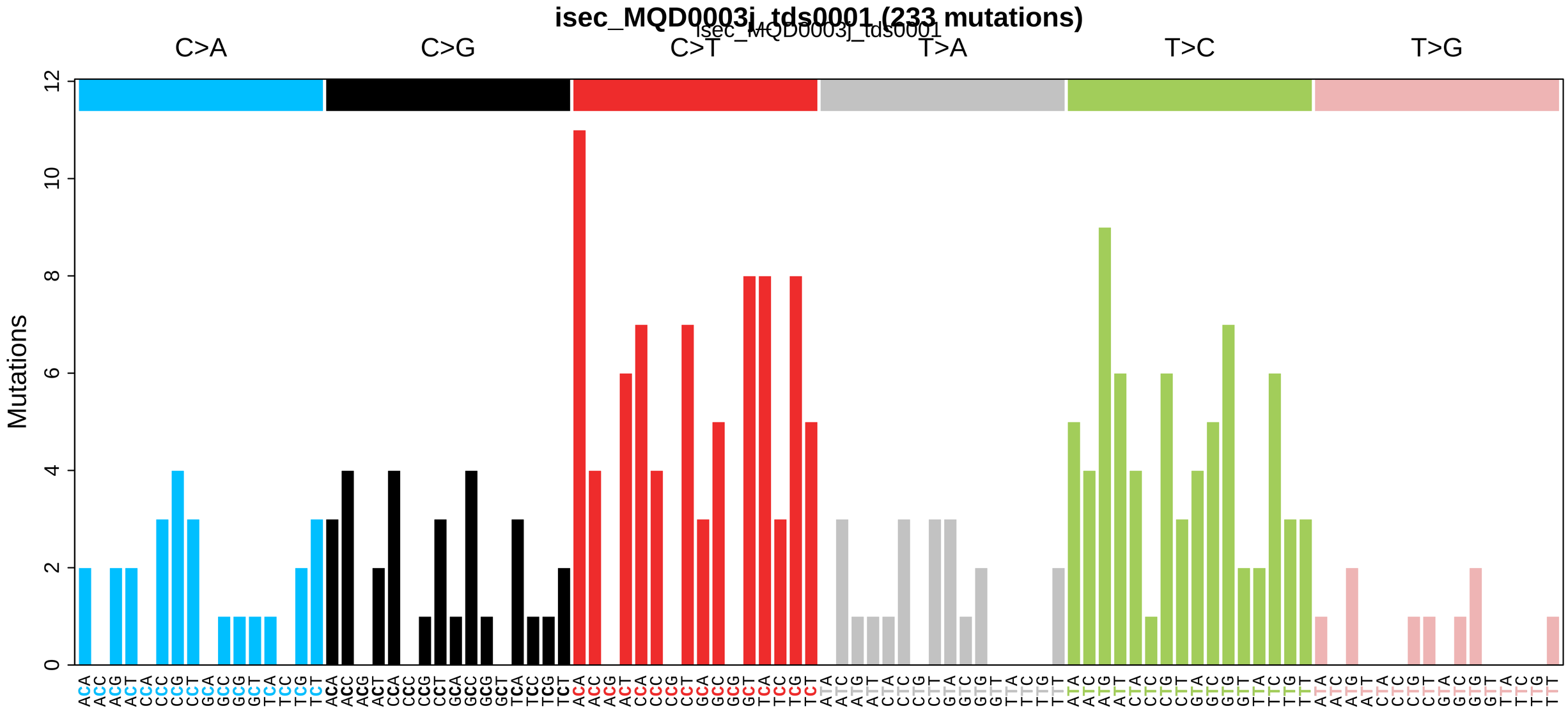

isec_MQD0002I_tds0001 (255 mutations)
isec_MQD0002I_tds0001
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
2
4
6
8
10
12
14

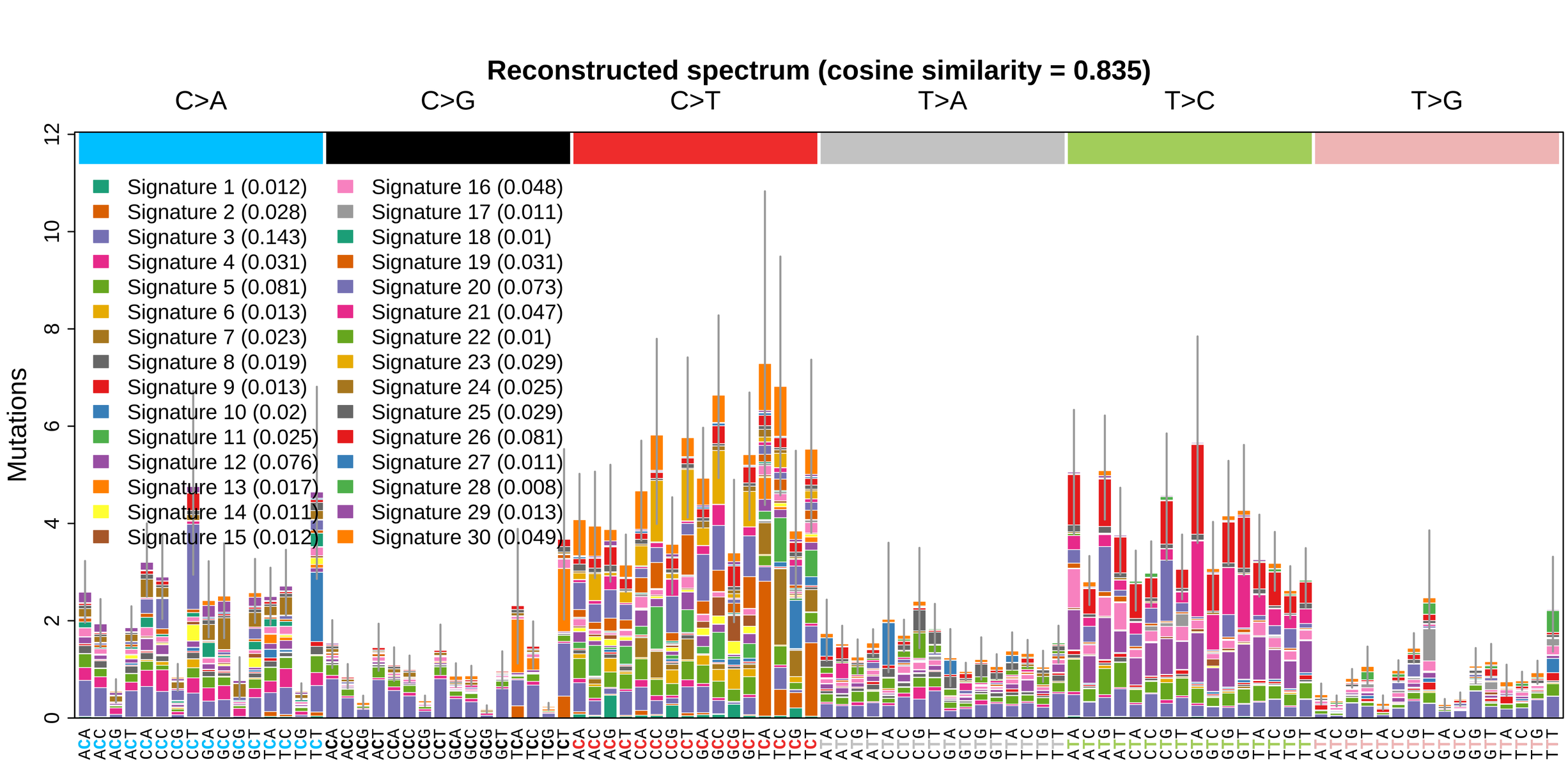

Reconstructed spectrum (cosine similarity = 0.866)
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
2
4
6
8
10
12
14
Signature 1 (0.01)
Signature 2 (0.03)
Signature 3 (0.04)
Signature 4 (0.023)
Signature 5 (0.043)
Signature 6 (0.01)
Signature 7 (0.03)
Signature 8 (0.025)
Signature 9 (0.022)
Signature 10 (0.011)
Signature 11 (0.055)
Signature 12 (0.217)
Signature 13 (0.007)
Signature 14 (0.01)
Signature 15 (0.011)
Signature 16 (0.048)
Signature 17 (0.017)
Signature 18 (0.009)
Signature 19 (0.028)
Signature 20 (0.029)
Signature 21 (0.04)
Signature 22 (0.011)
Signature 23 (0.025)
Signature 24 (0.014)
Signature 25 (0.03)
Signature 26 (0.111)
Signature 27 (0.008)
Signature 28 (0.013)
Signature 29 (0.011)
Signature 30 (0.061)


isec_MQD0003d_tds0001 (530 mutations)
isec_MQD0003d_tds0001
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
5
10
15
20
25
Reconstructed spectrum (cosine similarity = 0.913)
Signature 1 (0.006)
Signature 2 (0.017)
Signature 3 (0.185)
Signature 4 (0.033)
Signature 5 (0.075)
Signature 6 (0.005)
Signature 7 (0.01)
Signature 8 (0.015)
Signature 9 (0.027)
Signature 10 (0.024)
Signature 11 (0.012)
Signature 12 (0.057)
Signature 13 (0.007)
Signature 14 (0.005)
Signature 15 (0.005)
Signature 16 (0.259)
Signature 17 (0.01)
Signature 18 (0.019)
Signature 19 (0.011)
Signature 20 (0.011)
Signature 21 (0.019)
Signature 22 (0.007)
Signature 23 (0.011)
Signature 24 (0.035)
Signature 25 (0.023)
Signature 26 (0.026)
Signature 27 (0.01)
Signature 28 (0.011)
Signature 29 (0.019)
Signature 30 (0.049)


isec_MQD0003e_tds0001 (337 mutations)
isec_MQD0003e_tds0001
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
5
10
15
Reconstructed spectrum (cosine similarity = 0.876)
Signature 1 (0.009)
Signature 2 (0.021)
Signature 3 (0.147)
Signature 4 (0.012)
Signature 5 (0.061)
Signature 6 (0.008)
Signature 7 (0.039)
Signature 8 (0.024)
Signature 9 (0.024)
Signature 10 (0.02)
Signature 11 (0.041)
Signature 12 (0.186)
Signature 13 (0.009)
Signature 14 (0.008)
Signature 15 (0.007)
Signature 16 (0.091)
Signature 17 (0.007)
Signature 18 (0.007)
Signature 19 (0.032)
Signature 20 (0.028)
Signature 21 (0.017)
Signature 22 (0.011)
Signature 23 (0.028)
Signature 24 (0.006)
Signature 25 (0.031)
Signature 26 (0.056)
Signature 27 (0.007)
Signature 28 (0.007)
Signature 29 (0.009)
Signature 30 (0.049)


isec_MQD0003g_tds0001 (223 mutations)
isec_MQD0003g_tds0001
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
5
10
15
Reconstructed spectrum (cosine similarity = 0.831)
Signature 1 (0.049)
Signature 2 (0.009)
Signature 3 (0.089)
Signature 4 (0.025)
Signature 5 (0.108)
Signature 6 (0.026)
Signature 7 (0.016)
Signature 8 (0.023)
Signature 9 (0.031)
Signature 10 (0.011)
Signature 11 (0.031)
Signature 12 (0.039)
Signature 13 (0.012)
Signature 14 (0.014)
Signature 15 (0.014)
Signature 16 (0.164)
Signature 17 (0.012)
Signature 18 (0.013)
Signature 19 (0.043)
Signature 20 (0.031)
Signature 21 (0.022)
Signature 22 (0.009)
Signature 23 (0.033)
Signature 24 (0.026)
Signature 25 (0.02)
Signature 26 (0.055)
Signature 27 (0.01)
Signature 28 (0.021)
Signature 29 (0.018)
Signature 30 (0.027)


isec_MQD0003j_tds0001 (233 mutations)
isec_MQD0003j_tds0001
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
2
4
6
8
10
12


Reconstructed spectrum (cosine similarity = 0.835)
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
2
4
6
8
10
12
Signature 1 (0.012)
Signature 2 (0.028)
Signature 3 (0.143)
Signature 4 (0.031)
Signature 5 (0.081)
Signature 6 (0.013)
Signature 7 (0.023)
Signature 8 (0.019)
Signature 9 (0.013)
Signature 10 (0.02)
Signature 11 (0.025)
Signature 12 (0.076)
Signature 13 (0.017)
Signature 14 (0.011)
Signature 15 (0.012)
Signature 16 (0.048)
Signature 17 (0.011)
Signature 18 (0.01)
Signature 19 (0.031)
Signature 20 (0.073)
Signature 21 (0.047)
Signature 22 (0.01)
Signature 23 (0.029)
Signature 24 (0.025)
Signature 25 (0.029)
Signature 26 (0.081)
Signature 27 (0.011)
Signature 28 (0.008)
Signature 29 (0.013)
Signature 30 (0.049)

isec_MQD0004d_tds0001 (921 mutations)

isec_MQD0004d_tds0001

C>A C>G C>T T>A T>C T>G

Mutations

0 10 20 30 40 50

ACA ACC ACG ACT CCA CCC CCG CCT GCA GCC GCG GCT TCA TCC TCG TCT
ACA ACC ACG ACT CCA CCC CCG CCT GCA GCC GCG GCT TCA TCC TCG TCT
ACA ACC ACG ACT CCA CCC CCG CCT GCA GCC GCG GCT TCA TCC TCG TCT
ATA ATC ATG ATT CTA CTC CTG CTT GTA GTC GTG GTT TTA TTC TTG TTT
ATA ATC ATG ATT CTA CTC CTG CTT GTA GTC GTG GTT TTA TTC TTG TTT
ATA ATC ATG ATT CTA CTC CTG CTT GTA GTC GTG GTT TTA TTC TTG TTT

Reconstructed spectrum (cosine similarity = 0.886)

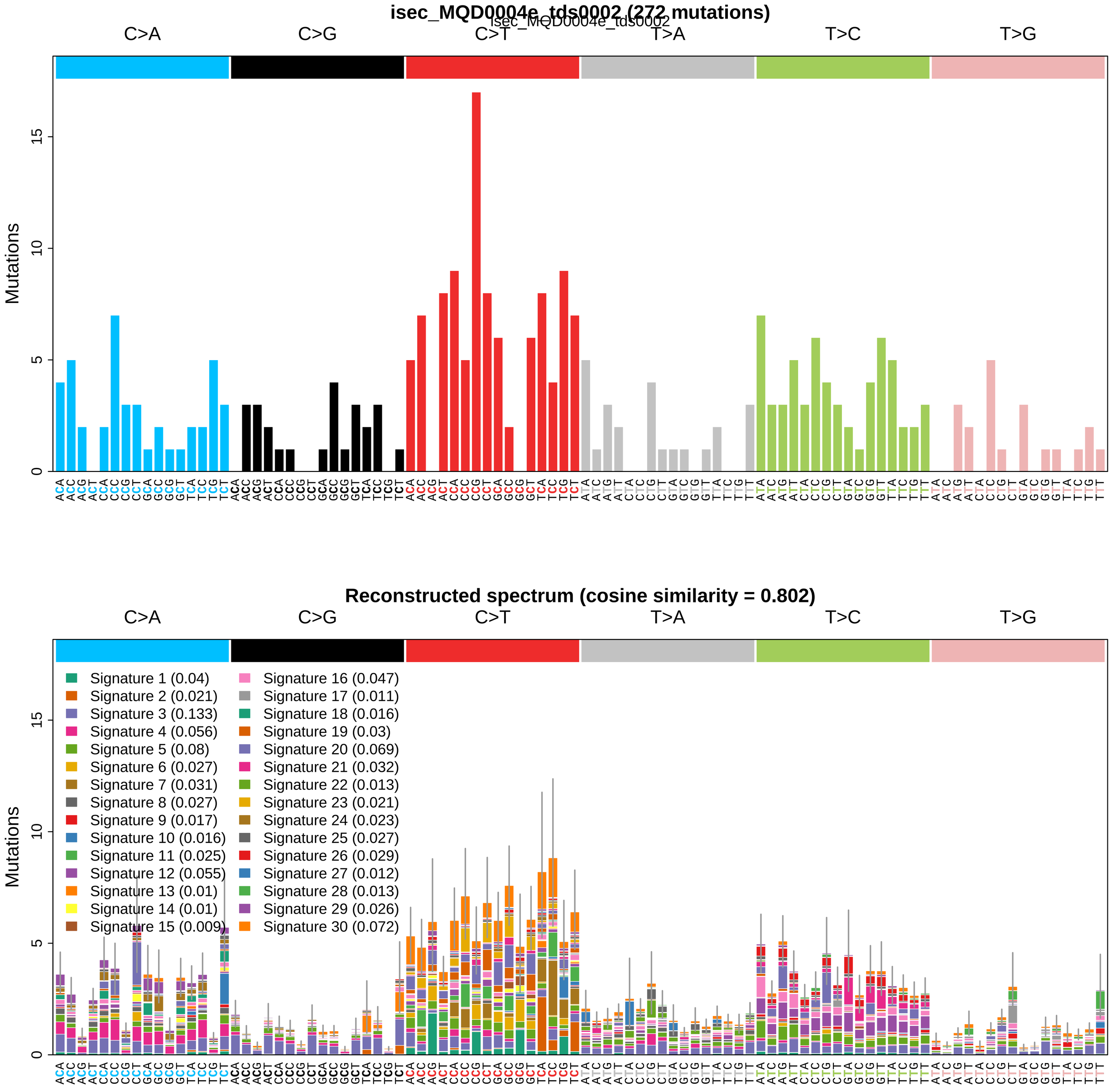
C>A C>G C>T T>A T>C T>G

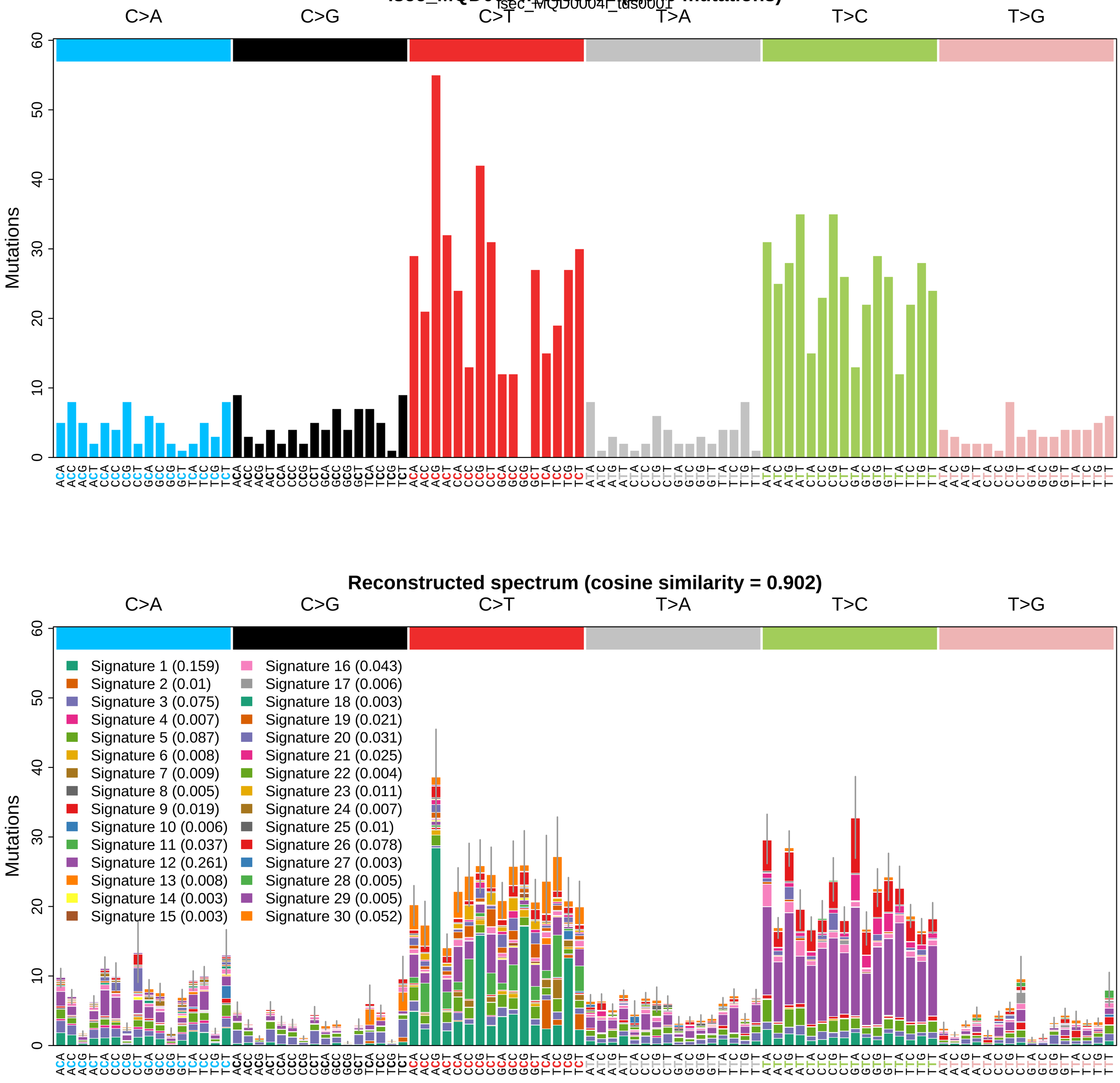
Signature 1 (0.004)
Signature 2 (0.007)
Signature 3 (0.144)
Signature 4 (0.004)
Signature 5 (0.062)
Signature 6 (0.004)
Signature 7 (0.03)
Signature 8 (0.004)
Signature 9 (0.012)
Signature 10 (0.013)
Signature 11 (0.034)
Signature 12 (0.289)
Signature 13 (0.009)
Signature 14 (0.003)
Signature 15 (0.003)
Signature 16 (0.062)
Signature 17 (0.008)
Signature 18 (0.002)
Signature 19 (0.016)
Signature 20 (0.01)
Signature 21 (0.028)
Signature 22 (0.006)
Signature 23 (0.035)
Signature 24 (0.003)
Signature 25 (0.01)
Signature 26 (0.099)
Signature 27 (0.003)
Signature 28 (0.022)
Signature 29 (0.002)
Signature 30 (0.073)

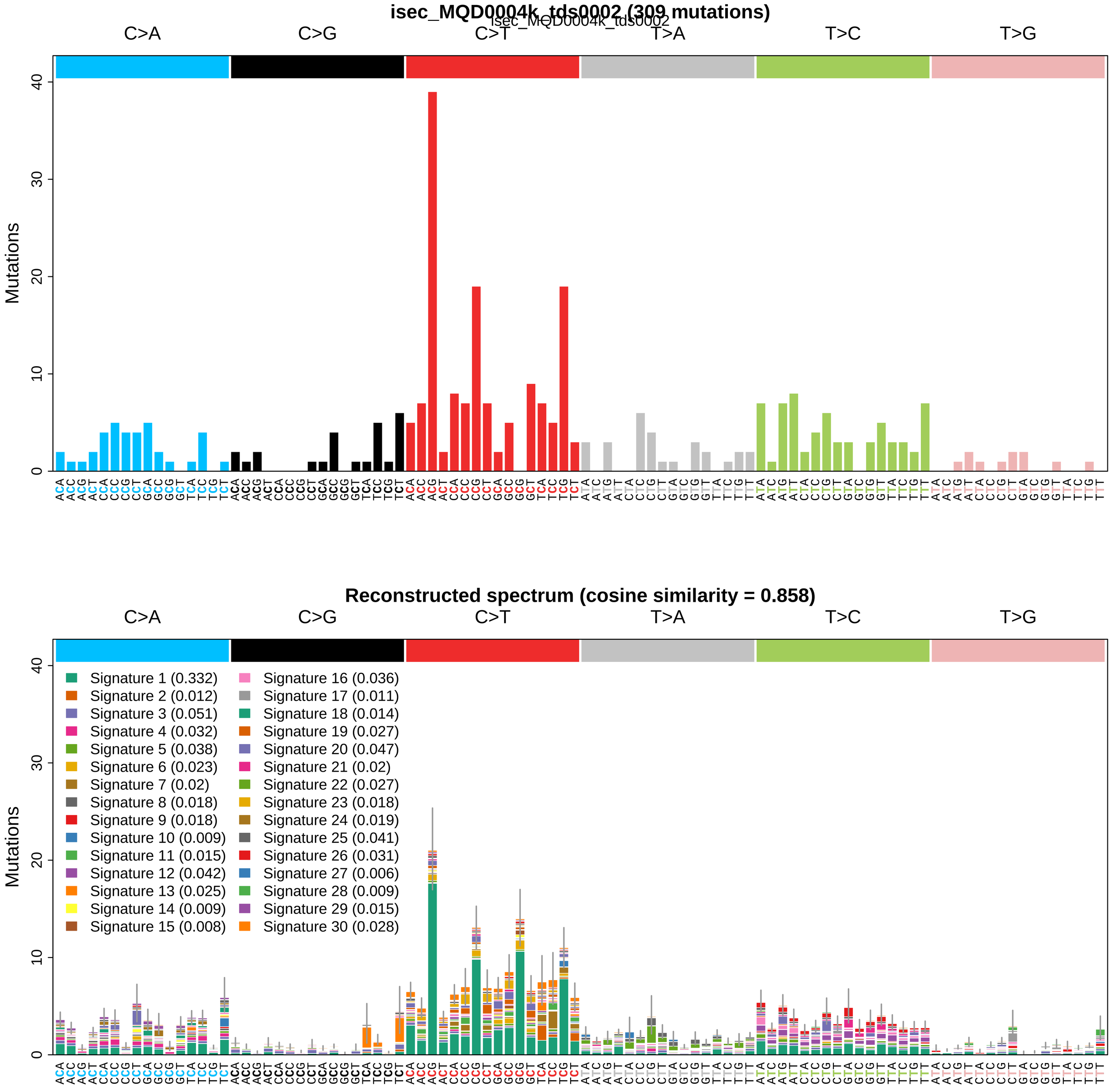
Mutations

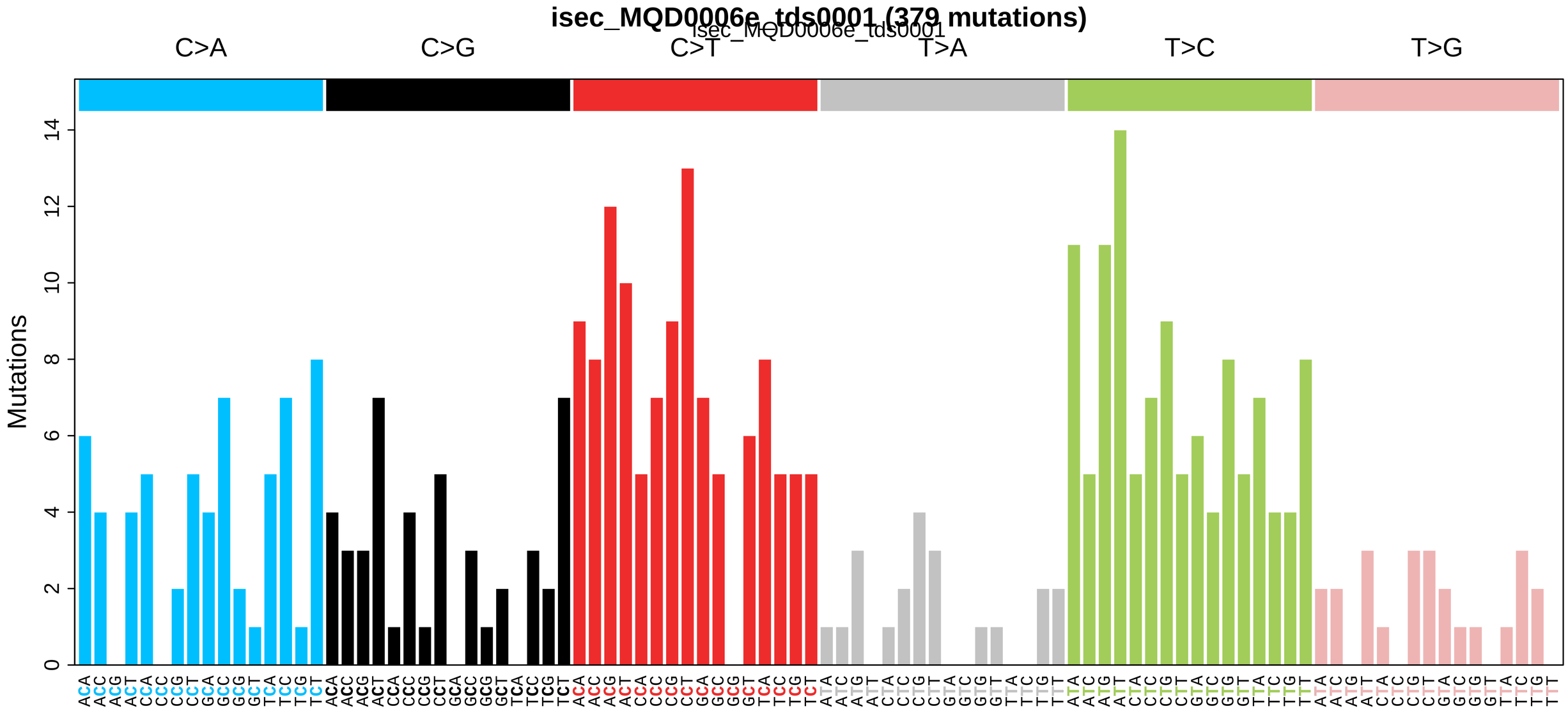
0 10 20 30 40 50

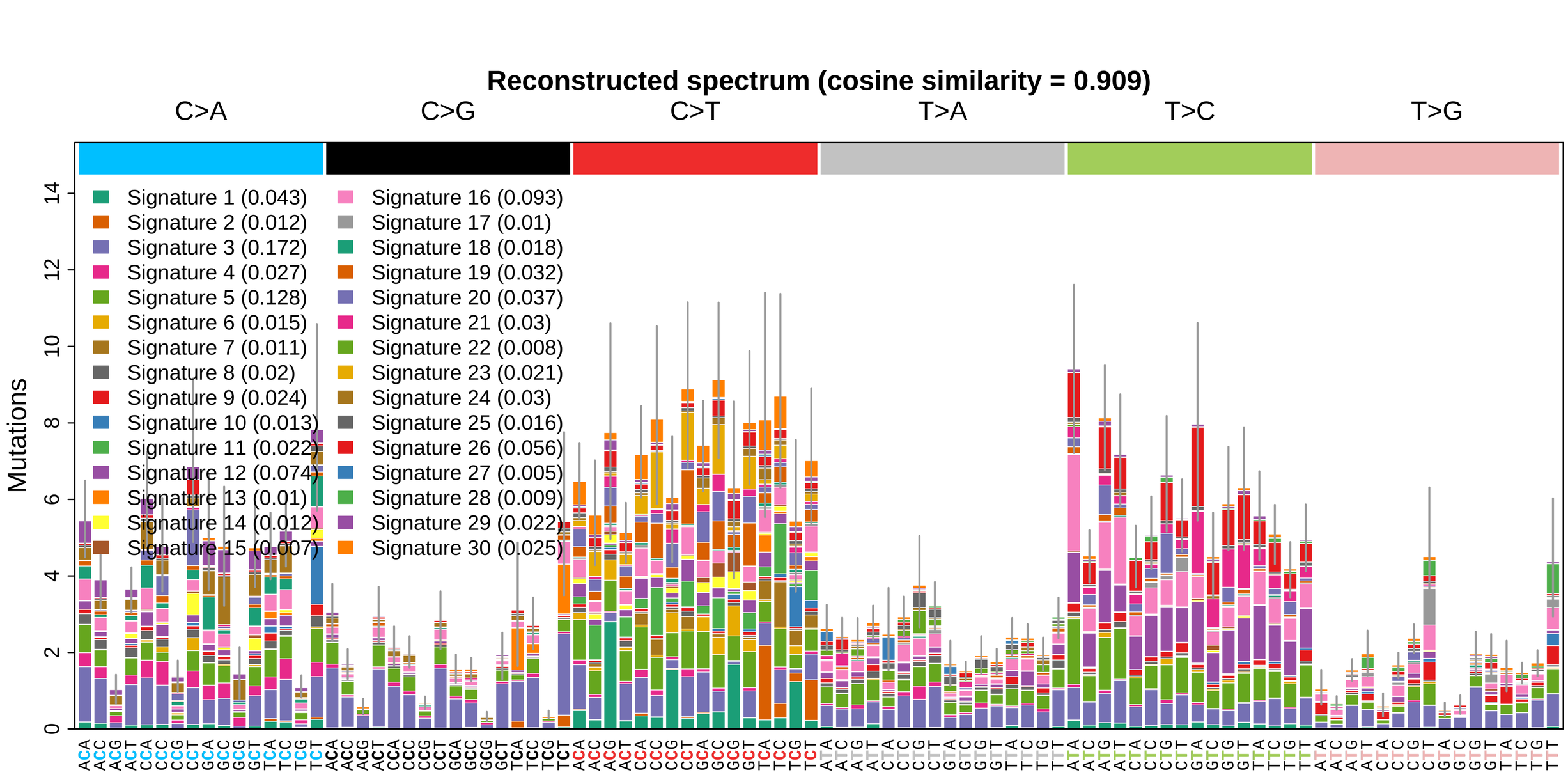
ACA ACC ACG ACT CCA CCC CCG CCT GCA GCC GCG GCT TCA TCC TCG TCT
ACA ACC ACG ACT CCA CCC CCG CCT GCA GCC GCG GCT TCA TCC TCG TCT
ACA ACC ACG ACT CCA CCC CCG CCT GCA GCC GCG GCT TCA TCC TCG TCT
ATA ATC ATG ATT CTA CTC CTG CTT GTA GTC GTG GTT TTA TTC TTG TTT
ATA ATC ATG ATT CTA CTC CTG CTT GTA GTC GTG GTT TTA TTC TTG TTT
ATA ATC ATG ATT CTA CTC CTG CTT GTA GTC GTG GTT TTA TTC TTG TTT


isec_MQD0004e_tds0002 (272 mutations)
isec_MQD0004e_tds0002
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
5
10
15
Reconstructed spectrum (cosine similarity = 0.802)
Signature 1 (0.04)
Signature 2 (0.021)
Signature 3 (0.133)
Signature 4 (0.056)
Signature 5 (0.08)
Signature 6 (0.027)
Signature 7 (0.031)
Signature 8 (0.027)
Signature 9 (0.017)
Signature 10 (0.016)
Signature 11 (0.025)
Signature 12 (0.055)
Signature 13 (0.01)
Signature 14 (0.01)
Signature 15 (0.009)
Signature 16 (0.047)
Signature 17 (0.011)
Signature 18 (0.016)
Signature 19 (0.03)
Signature 20 (0.069)
Signature 21 (0.032)
Signature 22 (0.013)
Signature 23 (0.021)
Signature 24 (0.023)
Signature 25 (0.027)
Signature 26 (0.029)
Signature 27 (0.012)
Signature 28 (0.013)
Signature 29 (0.026)
Signature 30 (0.072)


isec_MQD0004i_tds0001 (1,040 mutations)
isec_MQD0004i_tds0001
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
10
20
30
40
50
60
Reconstructed spectrum (cosine similarity = 0.902)
Signature 1 (0.159)
Signature 2 (0.01)
Signature 3 (0.075)
Signature 4 (0.007)
Signature 5 (0.087)
Signature 6 (0.008)
Signature 7 (0.009)
Signature 8 (0.005)
Signature 9 (0.019)
Signature 10 (0.006)
Signature 11 (0.037)
Signature 12 (0.261)
Signature 13 (0.008)
Signature 14 (0.003)
Signature 15 (0.003)
Signature 16 (0.043)
Signature 17 (0.006)
Signature 18 (0.003)
Signature 19 (0.021)
Signature 20 (0.031)
Signature 21 (0.025)
Signature 22 (0.004)
Signature 23 (0.011)
Signature 24 (0.007)
Signature 25 (0.01)
Signature 26 (0.078)
Signature 27 (0.003)
Signature 28 (0.005)
Signature 29 (0.005)
Signature 30 (0.052)


isec_MQD0004k_tds0002 (309 mutations)
isec_MQD0004k_tds0002
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
10
20
30
40
Reconstructed spectrum (cosine similarity = 0.858)
Signature 1 (0.332)
Signature 2 (0.012)
Signature 3 (0.051)
Signature 4 (0.032)
Signature 5 (0.038)
Signature 6 (0.023)
Signature 7 (0.02)
Signature 8 (0.018)
Signature 9 (0.018)
Signature 10 (0.009)
Signature 11 (0.015)
Signature 12 (0.042)
Signature 13 (0.025)
Signature 14 (0.009)
Signature 15 (0.008)
Signature 16 (0.036)
Signature 17 (0.011)
Signature 18 (0.014)
Signature 19 (0.027)
Signature 20 (0.047)
Signature 21 (0.02)
Signature 22 (0.027)
Signature 23 (0.018)
Signature 24 (0.019)
Signature 25 (0.041)
Signature 26 (0.031)
Signature 27 (0.006)
Signature 28 (0.009)
Signature 29 (0.015)
Signature 30 (0.028)


isec_MQD0006e_tds0001 (379 mutations)
isec_MQD0006e_tds0001
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
2
4
6
8
10
12
14


Reconstructed spectrum (cosine similarity = 0.909)
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
2
4
6
8
10
12
14
Signature 1 (0.043)
Signature 2 (0.012)
Signature 3 (0.172)
Signature 4 (0.027)
Signature 5 (0.128)
Signature 6 (0.015)
Signature 7 (0.011)
Signature 8 (0.02)
Signature 9 (0.024)
Signature 10 (0.013)
Signature 11 (0.022)
Signature 12 (0.074)
Signature 13 (0.01)
Signature 14 (0.012)
Signature 15 (0.007)
Signature 16 (0.093)
Signature 17 (0.01)
Signature 18 (0.018)
Signature 19 (0.032)
Signature 20 (0.037)
Signature 21 (0.03)
Signature 22 (0.008)
Signature 23 (0.021)
Signature 24 (0.03)
Signature 25 (0.016)
Signature 26 (0.056)
Signature 27 (0.005)
Signature 28 (0.009)
Signature 29 (0.022)
Signature 30 (0.025)

isec_MQD0006f_tds0001 (352 mutations)

isec_MQD0006f_tds0001

C>A C>G C>T T>A T>C T>G

Mutations

0 5 10 15 20

ACA ACC ACG ACT CCA CCC CCG CCT GCA GCC GCG GCT TCA TCC TCG TCT ACA ACC ACG ACT CCA CCC CCG CCT GCA GCC GCG GCT TCA TCC TCG TCT ACA ACC ACG ACT CCA CCC CCG CCT GCA GCC GCG GCT TCA TCC TCG TCT ATA ATC ATG ATT CTA CTC CTG CTT GTA GTC GTG GTT TTA TTC TTG TTT ATA ATC ATG ATT CTA CTC CTG CTT GTA GTC GTG GTT TTA TTC TTG TTT ATA ATC ATG ATT CTA CTC CTG CTT GTA GTC GTG GTT TTA TTC TTG TTT

Reconstructed spectrum (cosine similarity = 0.894)

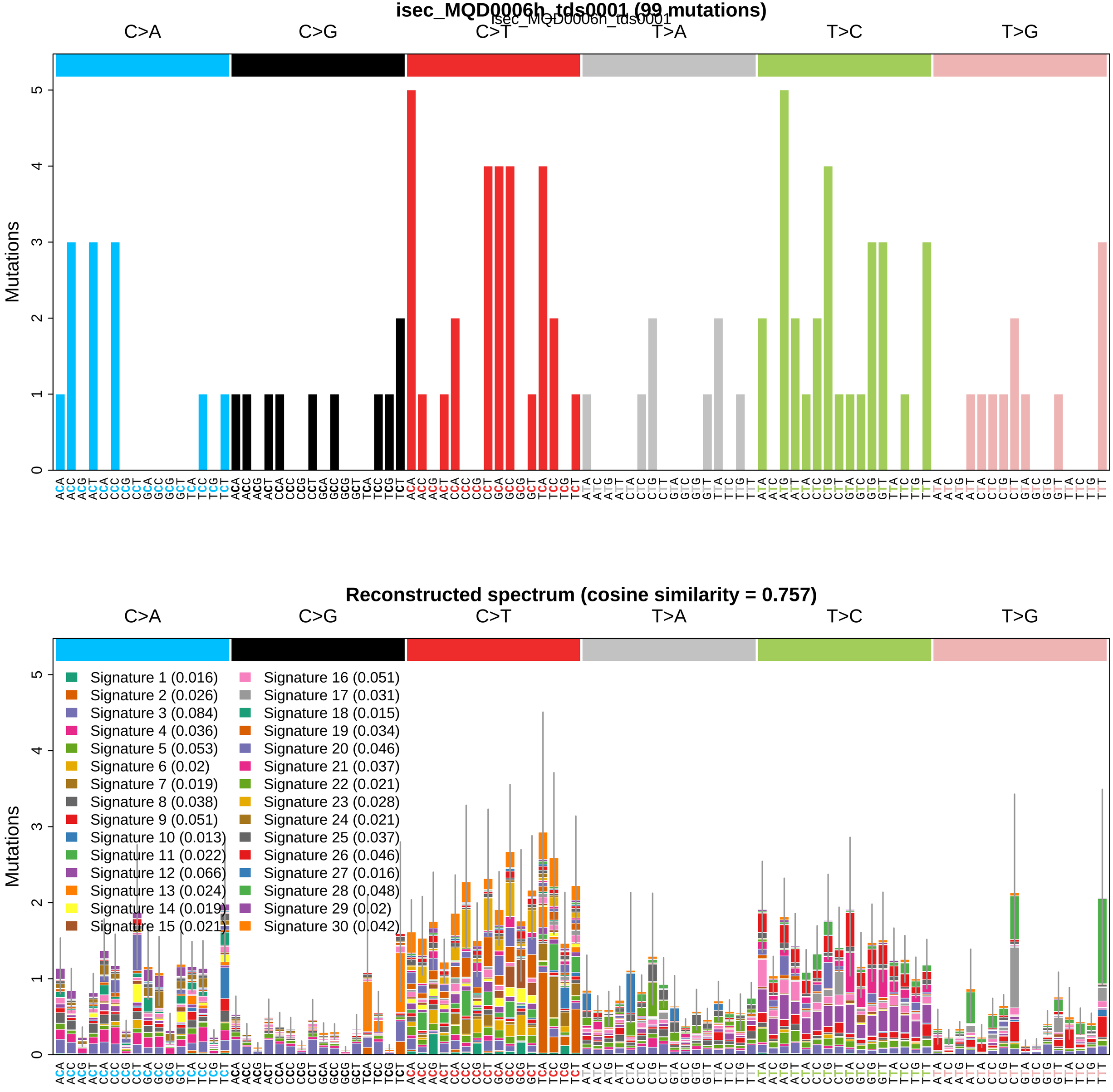
C>A C>G C>T T>A T>C T>G

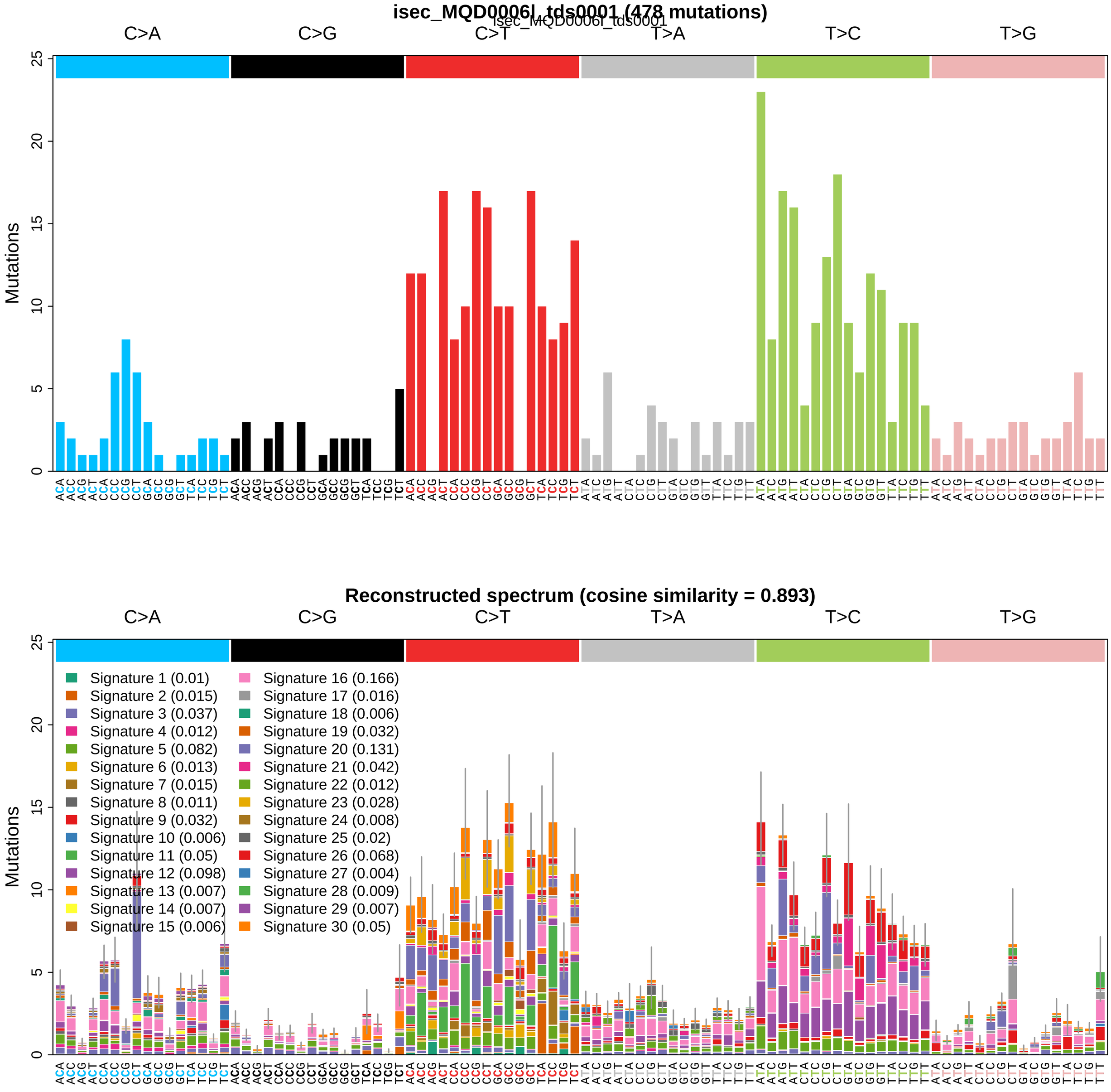
Mutations

0 5 10 15 20

Signature 1 (0.093)
Signature 2 (0.023)
Signature 3 (0.108)
Signature 4 (0.025)
Signature 5 (0.068)
Signature 6 (0.015)
Signature 7 (0.015)
Signature 8 (0.033)
Signature 9 (0.069)
Signature 10 (0.012)
Signature 11 (0.038)
Signature 12 (0.051)
Signature 13 (0.039)
Signature 14 (0.009)
Signature 15 (0.01)
Signature 16 (0.06)
Signature 17 (0.008)
Signature 18 (0.01)
Signature 19 (0.03)
Signature 20 (0.019)
Signature 21 (0.057)
Signature 22 (0.016)
Signature 23 (0.025)
Signature 24 (0.013)
Signature 25 (0.025)
Signature 26 (0.037)
Signature 27 (0.007)
Signature 28 (0.01)
Signature 29 (0.014)
Signature 30 (0.063)

ACA ACC ACG ACT CCA CCC CCG CCT GCA GCC GCG GCT TCA TCC TCG TCT ACA ACC ACG ACT CCA CCC CCG CCT GCA GCC GCG GCT TCA TCC TCG TCT ACA ACC ACG ACT CCA CCC CCG CCT GCA GCC GCG GCT TCA TCC TCG TCT ATA ATC ATG ATT CTA CTC CTG CTT GTA GTC GTG GTT TTA TTC TTG TTT ATA ATC ATG ATT CTA CTC CTG CTT GTA GTC GTG GTT TTA TTC TTG TTT ATA ATC ATG ATT CTA CTC CTG CTT GTA GTC GTG GTT TTA TTC TTG TTT


isec_MQD0006h_tds0001 (99 mutations)
isec_MQD0006h_tds0001
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
1
2
3
4
5
Reconstructed spectrum (cosine similarity = 0.757)
Signature 1 (0.016)
Signature 2 (0.026)
Signature 3 (0.084)
Signature 4 (0.036)
Signature 5 (0.053)
Signature 6 (0.02)
Signature 7 (0.019)
Signature 8 (0.038)
Signature 9 (0.051)
Signature 10 (0.013)
Signature 11 (0.022)
Signature 12 (0.066)
Signature 13 (0.024)
Signature 14 (0.019)
Signature 15 (0.021)
Signature 16 (0.051)
Signature 17 (0.031)
Signature 18 (0.015)
Signature 19 (0.034)
Signature 20 (0.046)
Signature 21 (0.037)
Signature 22 (0.021)
Signature 23 (0.028)
Signature 24 (0.021)
Signature 25 (0.037)
Signature 26 (0.046)
Signature 27 (0.016)
Signature 28 (0.048)
Signature 29 (0.02)
Signature 30 (0.042)


isec_MQD0006l_tds0001 (478 mutations)
isec_MQD0006l_tds0001
C>A
C>G
C>T
T>A
T>C
T>G
Mutations
0
5
10
15
20
25
Reconstructed spectrum (cosine similarity = 0.893)
Signature 1 (0.01)
Signature 2 (0.015)
Signature 3 (0.037)
Signature 4 (0.012)
Signature 5 (0.082)
Signature 6 (0.013)
Signature 7 (0.015)
Signature 8 (0.011)
Signature 9 (0.032)
Signature 10 (0.006)
Signature 11 (0.05)
Signature 12 (0.098)
Signature 13 (0.007)
Signature 14 (0.007)
Signature 15 (0.006)
Signature 16 (0.166)
Signature 17 (0.016)
Signature 18 (0.006)
Signature 19 (0.032)
Signature 20 (0.131)
Signature 21 (0.042)
Signature 22 (0.012)
Signature 23 (0.028)
Signature 24 (0.008)
Signature 25 (0.02)
Signature 26 (0.068)
Signature 27 (0.004)
Signature 28 (0.009)
Signature 29 (0.007)
Signature 30 (0.05)